Price Risk, Irreversible Investment, and the Welfare Case for the Section 45Z Credit

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One question, three results

Question. Section 45Z pays SAF producers a flat \$1.30–\$1.45 per gallon. The statute justifies it by carbon. The informal case adds “de-risking” irreversible investment. How much of the credit does each argument explain, and is a *flat* credit the right instrument for the risk part?

1. **Price risk justifies at most \$1.04/gal**, and \$0.52/gal at the midpoint risk weight. Risk alone cannot cover the credit.
2. **Carbon plus risk rationalize the credit** at an implied social cost of carbon of \$77–\$91/tCO_2, between the two federal values. The post-2025 rate prices carbon below both.
3. **The flat credit is the most expensive instrument** for the risk part: a price floor delivers the same capacity at 69% of the outlay, a two-sided contract at zero.

Roadmap: setting → framework and closed form → data → the three results → scope.

The setting: a supply chain that must be built before prices are known

U.S. SAF Grand Challenge: 3 B gal/yr of sustainable aviation fuel by 2030, 35 B by 2050.

The southeastern United States is the platform most often discussed for forest-residue SAF:

- 12 states, 1,374 counties; hardwood-residue pool 72.5 M dt/yr
- Pre-processing depots and gasification–Fischer–Tropsch biorefineries
- Capital is lumpy and irreversible; refineries live 20 years

The revenue price is uncertain. A producer is paid a drop-in fuel price plus the Renewable Fuel Standard credits the fuel earns. Weekly, 2013–2021, that price ranged over \$3.81–\$9.12/gal.

> Depot and refinery capacity must be committed before any of that price is observed. SAF deployment is a problem of investment under uncertainty.

The policy: Section 45Z pays by carbon intensity

$$\text{credit per gallon} \;=\; \text{base rate} \times \frac{50 - \mathrm{CI}}{50}$$

where CI is the pathway's life-cycle carbon intensity in $\mathrm{kgCO_2e/mmBTU}$.

- As enacted (IRA 2022): SAF base rate \$1.75/gal $\Rightarrow$ \$1.30–\$1.45/gal for forestry-residue FT (CI $\approx$ 8.6–12.9)
- 2025 reconciliation act: base rate aligned to \$1.00/gal after 2025 $\Rightarrow$ \$0.74–\$0.83/gal

Two justifications in circulation

1. **Carbon.** The statute's own logic: pay for avoided emissions.
2. **De-risking.** Irreversible capacity against volatile fuel prices. Never priced.

We price both in one framework, show they add exactly, and then ask whether a flat per-gallon payment is the right way to deliver the risk component at all.

What we do

Empirical model

- Social planner, make-or-buy: domestic SAF vs. imported fossil jet at \$3.50/gal
- Two-stage stochastic MILP, 1,374 counties, 20 price scenarios
- Price distribution: kernel density on a constructed drop-in price, 2013–2021

Closed-form benchmark

- Two-state reduction calibrated to the same distribution
- Theorem: the risk-premium subsidy $\sigma^* = \lambda\,\pi_H\,(w_H - w_L)$
- Prop. 1: carbon and risk components add exactly
- Prop. 2: expected outlay $T_{\text{flat}} > T_{\text{floor}} > T_{\text{CfD}} = 0$

Where we sit. Bioenergy siting MILPs (Eksioglu et al. 2009; Marvin et al. 2012) have the space but no welfare or true uncertainty; CVaR stochastic programming (Rockafellar–Uryasev 2000) has the risk machinery but not the landscape; instrument comparisons under price risk (Boomsma et al. 2012) were never tied to a statutory credit.

The planner's problem: make or buy

Output Q of SAF in scenario ω; consumer and producer surplus; unmet demand met by imported jet fuel:

$$\max_{\substack{y\in\{0,1\}^{|\mathcal{D}|+|\mathcal{R}|} \\ \mathbf{x},\mathbf{z},s\geq 0}} \mathbb{E}_\omega\left[\underbrace{\mathrm{CS}(\mathbf{z};\omega)}_{\text{consumers}} + \underbrace{\mathrm{PS}(\mathbf{x},\mathbf{z},y;\omega)}_{\text{revenue minus chain cost}} - \underbrace{p^{\text{jet}}\, s(\omega)}_{\text{slack imports at \$3.50}}\right]$$

Timing

1. **Here-and-now:** choose depots and refineries y before the price is known
2. Nature draws ω: the SAF price $p_{\text{SAF}}(\omega)$ (20 KDE bins) and a feedstock-quality shock
3. **Recourse:** choose residue flows $\mathbf{x}(\omega), \mathbf{z}(\omega)$ and imports $s(\omega)$ after ω is seen

The outside option prices unmet demand. Without it, not building is free and the build-versus-no-build margin is undefined. The carbon price is held at zero inside the program so that the price-risk channel is isolated; carbon is priced analytically later.

Risk aversion: whose, and over what

Replace the expectation with a mean–CVaR functional:

$$(1-\lambda)\,\mathbb{E}[W] + \lambda\ \mathrm{CVaR}_\alpha[W], \qquad \lambda \in [0,1],\ \ \alpha = 0.95$$

$\mathrm{CVaR}_{0.95}[W]$ is the average welfare in the worst 5% of scenarios; λ is the weight on that tail.

Whose risk? The **committing agent**: the Southeast regional aggregate, or the non-diversified investor whose balance sheet carries the irreversible capacity.
Who pays the correction? The **external financier**: the federal side, whose taxpayer pooling makes it risk-neutral (Arrow–Lind 1970).

> $\lambda > 0$ is not a social preference. It is the shadow price of missing markets: no ten-year hedge exists for the SAF premium, so the committing agent cannot sell its low-price exposure. σ^* prices a jurisdictional risk-bearing wedge, not a market failure in the Pigouvian sense.

The closed-form benchmark: a two-state economy

Reduce the program to one output decision Q with two demand states $\omega \in \{H, L\}$, probabilities (π_H, π_L):

- Inverse demand $P^d(Q; \omega) = w(\omega) - bQ$, intercepts $w_H > w_L > c$
- Marginal cost c; capacity cost $\frac{k}{2}Q^2$; combined curvature $K = b + k$
- Net premium $A(\omega) = w(\omega) - c$; per-state welfare $W(Q; \omega) = A(\omega)\,Q - \frac{K}{2}Q^2$

Two planners

- Risk-neutral optimum: $Q^*_{\mathrm{RN}} = \mathbb{E}[A]/K$
- Mean–CVaR optimum $Q^*_{\mathrm{risk}} < Q^*_{\mathrm{RN}}$: the tail-averse planner builds less

Risk-premium subsidy σ^*: the per-unit transfer that moves the mean–CVaR planner back to Q^*_{RN}. It is the "de-risking" argument, priced.

Theorem 1: the risk premium in three factors

Theorem 1. Under the maintained assumptions,

$$\sigma^*(\alpha) = \lambda\,(w_H - w_L)\,\min\left\{\tfrac{\alpha\,\pi_L}{1-\alpha},\ \pi_H\right\}$$

and for $\alpha \geq \pi_H$: $\boxed{\sigma^* = \lambda\,\pi_H\,(w_H - w_L)}$

$$\sigma^* = \underbrace{\lambda}_{\text{risk weight}} \times \underbrace{\pi_H}_{\text{upside probability}} \times \underbrace{(w_H - w_L)}_{\text{price spread}}$$

The baseline ($\alpha = 0.95$, $\pi_H = 0.472$) is in the second case: CVaR collapses to the low state's welfare level and σ^* no longer depends on α. No cost term, no curvature: only the price distribution and the risk weight matter. At $\lambda = 0$ it recovers Weitzman (1974).

Two propositions on top of the theorem

Proposition 1 (additivity). If each gallon carries an external benefit v (avoided emissions $\times$ carbon price) not already in the market price, the credit that restores first-best output is

$$\sigma^*_{\mathrm{tot}} = v + \sigma^*, \qquad \text{with no interaction term.}$$

The Pigouvian part can be priced off a life-cycle differential and a social cost of carbon without re-solving the program.

Proposition 2 (instrument form). Three instruments restore Q^*_{RN}; their expected per-unit fiscal outlays are strictly ranked for $\lambda > 0$:

$$0 = T_{\mathrm{CfD}} \ < \ T_{\mathrm{floor}} = \sigma^* \frac{\pi_L}{\pi_L + \lambda \pi_H} \ < \ T_{\mathrm{flat}} = \sigma^*.$$

A flat credit shifts the *mean* of the payoff distribution; the distortion it corrects is a *variance* phenomenon.

Data: the price distribution

No deep SAF spot market exists. We construct the revenue price of a drop-in fuel producer:

slate-weighted wholesale gasoline–diesel price	mean \$3.10/gal
+ RFS credit value (D-code RIN price × 1.7)	mean \$2.96/gal
= constructed drop-in revenue price	mean \$5.98/gal, s.d. \$1.27/gal

Weekly, August 2013 to December 2021.

- Gaussian KDE, discretized into 20 equal-width bins on [\$3.81, \$9.12]; binned mean \$6.00
- The *level* is policy-inclusive (half of it is RIN value), so the credit we price is incremental to the existing RIN stack
- The *dispersion* is genuine producer risk: RIN volatility passes one-for-one into the margin

Other inputs. Residue pool 72.5 M dt/yr (Stage-1 LP); refinery capital \$947 M, operating cost \$1.96/gal net of co-products; demand floor 3.0 B gal/yr; outside option \$3.50/gal fossil jet, policy-free basis.

From 20 bins to two states

Split the KDE at the bin-8/9 boundary ($5.93/gal, nearest the mean):

	High state	Low state
Probability	$\pi_H = 0.472$	$\pi_L = 0.528$
Conditional mean price	$w_H = \$7.16/\text{gal}$	$w_L = \$4.97/\text{gal}$

Spread $w_H - w_L = \$2.19/\text{gal}$.

$$\sigma^* = \lambda \times 0.472 \times 2.19 = \lambda \times \$1.04/\text{gal} \qquad \Longrightarrow \qquad \sigma^*(0.5) \approx \$0.52/\text{gal}$$

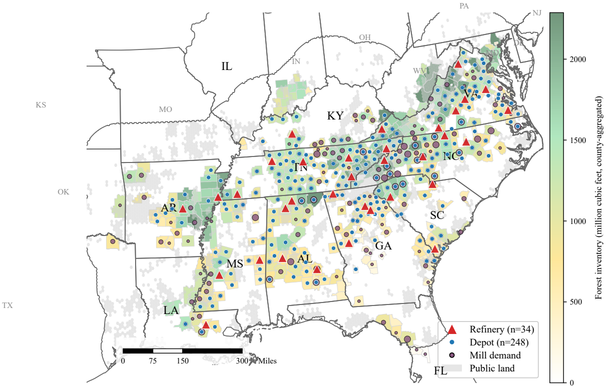
The bridge from realized price to demand intercept is a parallel shift: a price realization moves willingness to pay one-for-one. Robust to bin count (10 or 40 bins move σ^* by under 6%) and bandwidth (under 3%).

Results at a glance

Object	Value	Reading
Expected regret of committing early	\$299 M/yr	about a fifth of the chain's \$1.43 B/yr surplus
5%-tail of that regret	\$921 M/yr	3.1$\times$ the mean; all in low-price scenarios
Risk-premium subsidy σ^*	\$0.52/gal	at $\lambda = 0.5$; at most \$1.04 at any λ
Statutory credit	\$1.30–\$1.45/gal	risk alone cannot cover it
Implied social cost of carbon	\$77–\$91/tCO_2	between IWG \$51 and EPA \$190
Price floor at \$5.64/gal	69% of flat outlay	same capacity, \$0.48 B/yr saved
Two-sided CfD at \$6.00/gal	zero expected outlay	risk moves to the federal balance sheet

All flows single-year-equivalent in year-0 terms; capital annuitized over 20 years at 10%; 2022 USD.

Baseline: 248 depots, 34 refineries

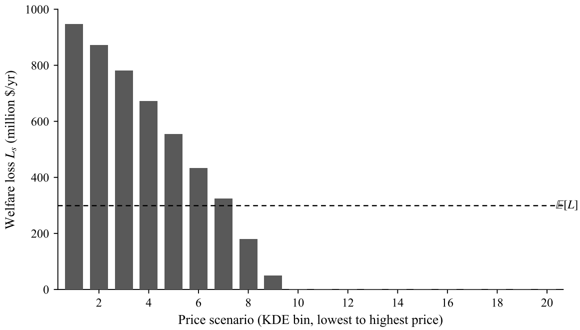


Deterministic solve at the central scenario ($6.05/gal):

- 248 depots, 34 refineries, 10 of 12 states
- 3.32 B gal/yr, demand floor met with zero imports
- Chain surplus (PVC) $1.43 B/yr
- Build-versus-import crossing at ≈$5.5/gal; expansion band $5.8–$6.6/gal

The plan is a feasible incumbent, not a certified optimum (18% MIP gap at the time limit). It is treated as a fixed commitment.

Result 1: the cost of committing early sits in the low-price tail

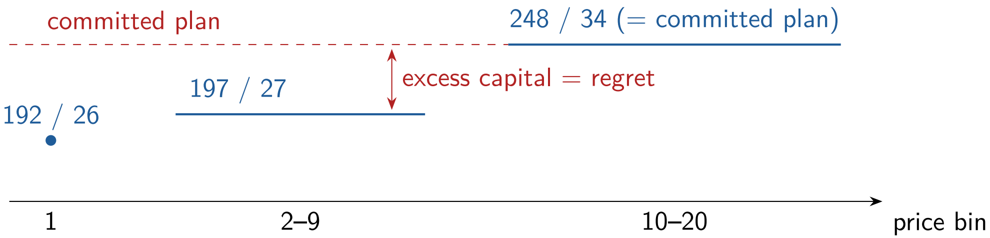


Regret in scenario s: what a planner who knew the price would have earned, minus what the committed plan earns.

- Losses only in the 9 lowest-price bins: \$947 M at bin 1 down to \$48 M at bin 9; zero above (41% of probability)
- $\mathbb{E}[L] = \$299 M/yr$
- $\mathrm{CVaR}_{0.95}[L] = \$921 M/yr$, 3.1× the mean

Wait-and-see values are certified against the candidate pool; residual gaps 1.3–3.9% make these lower bounds ($\mathbb{E}[L] \leq \$827$ M).

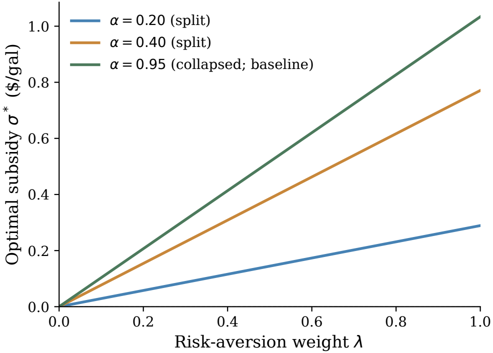
Why the tail: overbuilding against a low price is expensive



Scenario-optimal siting expands weakly in price: 192/26 depots and refineries at bin 1, 197/27 across bins 2–9, 248/34 from bin 10 up.

Overbuilding against a low price is costly; underbuilding against a high price is cheap at the margin, because extra residue comes from farther away and the consumer-surplus integral flattens. No scenario wants more than 248/34. The committed plan's welfare tail sits 33% below its mean, in the two lowest-price bins: that tail is what σ^* prices.

Result 2: price risk justifies at most $1.04/gal



$\sigma^*(\lambda)$ at the KDE calibration

$\sigma^*(\lambda) = \lambda \times \$1.04$/gal is linear in the risk weight:

- $\lambda = 0$: risk-neutral, no subsidy (Weitzman)
- $\lambda = 0.5$: $0.52/gal
- $\lambda = 1$: tail-only planner, $1.04/gal; empirical counterpart: the bin-1 plan, 192/26

> $\sigma^*(\lambda) \leq \$1.04 < \$1.30$ for every admissible λ. Price risk alone cannot cover the statutory credit, however tail-averse the planner.

Omitted channels: financing frictions raise σ^*; offtake coverage, cost–price co-movement, and hedging lower it (scope slide).

Does the 700k-variable integer model agree? Candidate evaluation

Hold six candidate plans fixed (248/34 and the low-price wait-and-see topologies) and evaluate each across all 20 bins:

	Committed 248/34	Contracted 197/27
Expected welfare $\mathbb{E}[W]$	**\$17.004 B/yr**	\$16.904 B/yr
Expected regret $\mathbb{E}[L]$	**\$300 M/yr**	\$400 M/yr
Regret tail $\mathrm{CVaR}_{0.95}[L]$	**\$931 M/yr**	\$1,837 M/yr
Welfare tail $\mathrm{CVaR}_{0.95}[W]$	lower	**higher by \$0.93 B/yr**

Expectation and regret favor the big plan; the welfare tail favors the plan one-fifth smaller. The mean–CVaR preference switches at $\lambda^* \approx 0.10$ (at $\lambda = 0.5$ the small plan wins by \$0.41 B/yr). A tail-averse planner builds one-fifth less: the contraction Theorem 1 prices, now on the integer chain. The joint mean–CVaR MILP itself is not solved; evaluation is exact on fixed plans.

Does the credit undo the distortion? The injection test

Inject the credit into the planner's program on the model's own two-state collapse, sitings fixed, LP-exact:

Configuration	Ranking	Wedge to leader
$\lambda = 0$	197/27 $\succ$ 248/34 $\succ$ 192/26	\$30.8 M/yr
$\lambda = 0.5$, no credit	248/34 falls to last	\$306.9 M/yr
$\lambda = 0.5 + \sigma^* = \$0.518$	risk-neutral order restored	\$47.8 M/yr

The credit removes 94% of the risk-induced distortion in the plan comparison. The MILP also pins down the closed form's two free parameters, K and c (backup).

Result 3a: carbon plus risk, and the implied social cost of carbon

The Pigouvian component. Every domestic gallon displaces one imported fossil gallon: $\Delta e = 10.2$ kgCO$_2$e/gal, so $v = \Delta e \times \text{SCC}$.

Federal SCC vintage	\$/tCO$_2$	v (\$/gal)
IWG interim (2021)	51	0.52
EPA (2023)	190	1.94

Carbon plus risk spans \$1.04–\$2.46/gal, a band that contains the statutory \$1.30–\$1.45.

Invert: what carbon price makes the credit exactly right?

$$\text{SCC}^{\text{implied}} = \frac{\text{credit} - \sigma^*}{\Delta e} = \frac{\$1.30\text{–}\$1.45 - \$0.52}{10.2\ \text{kg}} = \$77\text{–}\$91/\text{tCO}_2$$

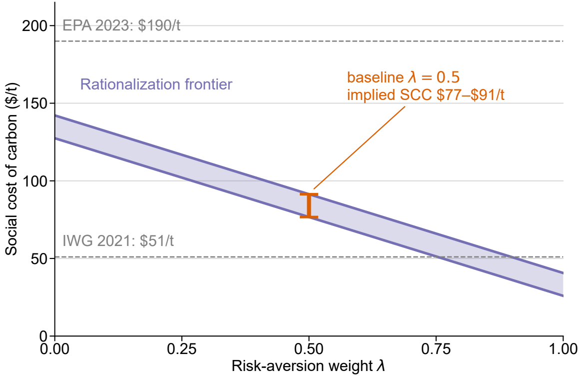
between the two federal vintages. A $\pm 20\%$ band on Δe keeps it there.

The implied SCC is a schedule, not a number

Reading	Assumption changed	$/t$CO_2$
Baseline (incremental)	as-enacted rate; flat credit; $\lambda = 0.5$	**77–91**
Instrument-efficient	risk paid via floor/CfD; flat credit prices carbon alone	127–142
Maximal risk aversion	$\lambda = 1$	26–41
Statutory emissions basis	statute's $(50 - \mathrm{CI})$ replaces physical Δe	169–180
Post-2025 rate	base \$1.00 $\Rightarrow$ credit \$0.74–\$0.83	22–30
Full RIN capitalization	embedded \$2.96/gal read as carbon support	≈367–382

"Is 45Z too generous?" has no vintage-independent answer. At the EPA value the carbon channel alone (\$1.94) exceeds the credit; at the IWG value the credit needs $\lambda \approx 0.75$–0.90. The news is the last-but-one row: **the rate now in force prices carbon below even the IWG floor.**

The rationalization frontier



Pairs (λ, SCC) at which carbon plus risk equals the statutory credit:

$$0.0102\,\mathsf{SCC} + 1.036\,\lambda \in [1.30, 1.45]$$

- Risk-neutral planner: needs SCC $127–$142
- Midpoint $\lambda = 0.5$: SCC $77–$91
- IWG $51: needs $\lambda \in [0.75, 0.90]$

The implied SCC is an internal-consistency price, not a revealed-preference estimate of what legislators believe carbon is worth.

Result 3b: the flat credit is the most expensive instrument for risk

The committing agent under-builds because the low-price tail is too costly to bear, not because expected returns are too low. A flat credit pays the same in high-price states, where no correction is needed, as in the low-price states that cause the distortion.

Instrument	Pays	$/gal expected	at 3.0 B gal/yr
Flat credit (45Z form)	$0.52 in every state	0.52	$1.55 B/yr
One-sided floor at $5.64	only when price $<$ $5.64	0.36	$1.07 B/yr
Two-sided CfD at $6.00	pays below, collects above the mean	0	0

- Floor: 69% of the flat outlay, $0.48 B/yr saved for the identical capacity; direct evaluation on all 20 bins gives $0.37/gal, within 4% of the closed form
- CfD: removes the variance itself; the federal side absorbs a symmetric exposure with s.d. $1.28/gal

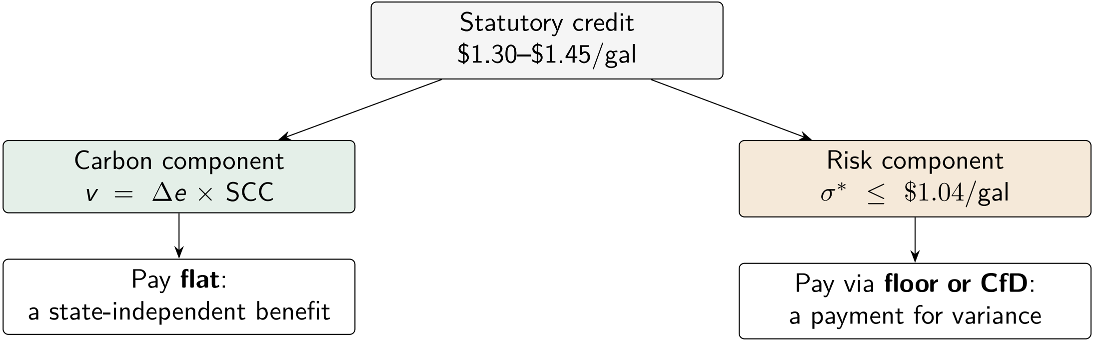
Qualifications, and what costly public funds add

Strike risk. A CfD at strike $\bar{w} + \delta$ costs δ per gallon in expectation: its edge over the floor vanishes at a \$0.36 error (6% of the mean). The floor's outlay rises at slope $\pi_L \approx 0.53$ per dollar of error. No deep SAF index exists to settle against.

Costly public funds ($\mu \in [1.10, 1.30]$, Dahlby 2008). The deadweight loss the risk correction recovers is $\sigma^{*2}/2K \approx$ \$31 M/yr (\$31–\$65 M across fits). Paying for it with a flat credit costs \$155–\$466 M/yr; with a floor, \$107–\$322 M/yr.

The flat credit fails a benefit–cost test for the risk component at any $\mu > 1.02$. Only the two-sided contract passes unambiguously.

Design implication: a mixed instrument



Re-packaging only the risk component as a floor holds planner output fixed and saves roughly \$0.5 B/yr; the U.K. Contracts-for-Difference scheme supplies the template (auction-discovered strikes, 15-year tenor, government-owned counterparty). The producer-side counterpart, through project finance, is the companion paper.

Scope: what partial equilibrium leaves out, with signs

Omitted channel	Effect on the decomposition
Stumpage-price endogeneity	moves levels, not σ^*; cost–price co-movement compresses the spread (natural hedge)
Land-use carbon debt	residue-specific debt stays inside the $\pm 20\%$ band; plantation-scale would not
LCFS rents, RIN pass-through	\$0.30/gal of stacked value raises the SCC needed to \$106–\$121
Financing frictions, hurdle wedges	raise σ^* (act on λ)
Offtake coverage η	$\sigma^*(\eta) = (1-\eta)\sigma^*$; $\eta = 0.5$ gives \$102–\$117

Even at full offtake coverage ($\sigma^* = 0$) the implied SCC is \$127–\$142/tCO_2, still between the federal vintages. The between-vintages reading survives any degree of coverage.

Limitations

1. **Single-period timing.** All capacity at $t = 0$; deferral value is bounded above by the value of information, which we estimate but do not certify.
2. **Exogenous mill demand.** Residue supply and pulpwood demand co-vary; a Faustmann-style response is left open.
3. **Asymmetric substitution.** SAF and fossil jet are substitutes for the buyer of last resort, not for the ESG-mandated offtaker.
4. **Risk-averse first stage not optimized.** Mean–CVaR evaluated exactly on six fixed plans; the joint MILP at intermediate λ is beyond the solver's tolerance for a 0.1–0.2% welfare effect.
5. **Price-state persistence.** Prices are treated as decade-scale states. Under full mean reversion the multi-year average price disperses far less and σ^* falls from \$0.52 toward \$0.11/gal. This is the main calibration risk on the risk side.

Takeaways

1. **Risk alone does not justify 45Z.** The risk-premium subsidy is \$0.52/gal at the midpoint and never exceeds \$1.04/gal; the credit is \$1.30–\$1.45. The integer chain reproduces the contraction, and injecting the credit closes 94% of the risk-induced wedge in the plan comparison.
2. **Carbon plus risk does, at a middling carbon price.** The as-enacted credit implies \$77–\$91/tCO_2, between the IWG and EPA values. The rate in force since 2025 implies \$22–\$30, below both.
3. **Pay less, and pay it differently.** The risk component is a payment for variance. A price floor delivers the same capacity at 69% of the flat credit's outlay; a two-sided contract at zero. Pay carbon flat; pay risk through the floor/CfD family.

Paper, code, and replication package available on request.

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Backup: proof sketch of Theorem 1

- With two states and $\alpha \geq \pi_H$, $\mathrm{CVaR}_\alpha[W] = W(Q; L)$: the tail is the low state.
- The mean–CVaR objective is $\mathcal{W}(Q) = A_{\text{eff}}\, Q - \frac{K}{2}Q^2$ with effective intercept $A_{\text{eff}} = (1-\lambda)\mathbb{E}[A] + \lambda A_L = \mathbb{E}[A] - \lambda \pi_H (w_H - w_L)$.
- Strictly concave $(-K)$, interior optimum $Q^*_{\text{risk}} = A_{\text{eff}}/K$.
- A per-unit credit σ enters like a price increment: $Q^*(\sigma) = (A_{\text{eff}} + \sigma)/K$. Setting $Q^*(\sigma) = \mathbb{E}[A]/K$ gives $\sigma^* = \lambda \pi_H (w_H - w_L)$.
- For $\alpha < \pi_H$ the tail straddles both states; the same algebra gives the $\alpha \pi_L/(1-\alpha)$ branch, continuous at $\alpha = \pi_H$.

Additivity: a state-independent v raises $\mathbb{E}[A]$ and A_L equally, so it moves the target without changing the wedge. Instrument ranking: the floor pays only in the low state, weight π_L; the CfD swaps the stochastic margin for the deterministic one at zero expected transfer.

Backup: the fixed-demand information-value accounting

The same information question on the fixed-demand chain-surplus (PVC) objective:

Quantity ($ M/yr)	Value
Incumbent-weighted wait-and-see expectation (lower bound on $\mathbb{E}[W^*]$)	1,541
Best candidate plan (lower bound on the recourse optimum)	1,286
EVPI, candidate-based estimate	255 (19.8%)
VSS	not reported

- Difference of two lower bounds: no determinate bias sign, so an *estimate*, not a certified bound
- Same order as the $299 M regret and the closed-form $190 M
- VSS would require a further nonzero-gap MILP without an anchoring bound; on the endogenous-demand candidate set it is zero, since the committed plan is the pool's expected-welfare argmax

Backup: calibrating K and c from the MILP

At interior optima the wait-and-see solves trace the closed form's first-order condition $w_\omega = c + K\,Q_\omega$. Regressing the per-bin demand intercept on realized wait-and-see output across the 20 bins:

- $K = 4.3 \times 10^{-9}$ \$/gal^2, $c = \$5.00$/gal ($R^2 = 0.68$); implied capacity curvature $k = K - b = 3.1 \times 10^{-10}$
- Subsample fits are unstable (bins 1–9: $K = 2.0 \times 10^{-9}$; bins 10–20: no sensible fit), evidence of integer kinks; carried as the calibration's uncertainty band
- Predicted bin-1 contraction $(w_{\text{bin1}} - c)/(\bar{w} - c) = 0.84$ vs. realized $192/248 = 0.77$: the MILP contracts more, consistent with minimum-throughput lumpiness
- Closed-form EVPI $\mathrm{Var}(A)/2K = \$190$ M/yr, the same order as the \$299 M regret and the \$255 M PVC estimate
- Recovered deadweight loss $\sigma^{*2}/2K = \$31$–\$65 M/yr, an order of magnitude below the flat credit's MCPF cost

Backup: solver profile and certification

- Stage-2 MILP: $\sim$700,000 continuous and $\sim$10,000 binary variables, $\sim$50,000 rows
- Per-scenario wait-and-see solves at a 5% gap target; realized gaps 1.3–3.9% after certification
- Certification: a candidate plan's fixed-siting value is feasible for the scenario's wait-and-see problem, so any candidate above the incumbent proves it suboptimal and triggers a warm-started re-solve
- Here-and-now values: fixed sitings, recourse solved to 10^{-12}
- Cross-plan statistics on an 81-breakpoint SOS2 grid; the 21-breakpoint grid's cross-plan margins were chord-error artifacts (sign reversed on refinement)
- Gap-implied allowance bounds $\mathbb{E}[L] \in [\$299, \$827]$ M/yr and the tail-to-mean ratio in $[1.1, 4.1]$

Backup: biorefinery cost against public design cases

Study	Pathway	TCI ($M)	Output (M gal/yr)	TCI/gal-yr
This study	residue FT, first-of-a-kind	947	50–100	9.5–19
Tan et al. (2015)	indirect liquefaction, *n*th plant	437.5	47.0	9.31
Davis et al. (2018)	biochemical, *n*th plant	697–758	31–33 (GGE)	22–23

Our $11/gal-yr is a 1.2× premium over the thermochemical *n*th-plant anchor, conservative against the 1.5–2.5× first-of-a-kind multipliers. Operating cost $1.96/gal net of co-products; sensitivity at $2.30 and $2.70 (hydrogen-inclusive): the 248/34 topology holds in 8 of 9 elasticity × cost cells. σ^* is invariant to all of this by construction.

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